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Paralel: 2

ANALISIS REGRESI

TUGAS 1

1. Data yang digunakan

Rata Rata Pengeluaran makanan sebulan (Variabel Y)	Rata-rata Pengeluaran Perkapita Sebulan (Variabel X)
4375285.714	1181394.841
2930571.429	1629940.476
5240785.714	6196726.19
3047742.857	1115171.131
3462857.143	1856422.619
6088701.429	1595075.238
2957571.429	861347.619
3762428.571	1162035.714
2821328.571	988615.4762
4184142.857	977857.1429
1257428.571	1023755.952
2495142.857	704411.9048
1322142.857	791019.8413
1518728.571	973036.3095
1816457.143	1111263.492
2748857.143	1159714.286
715414.2857	1137664.286
4803428.571	2251627.976
5149285.714	5316517.857
4375714.286	2782543.651
685714.2857	1302964.286
4348285.714	1416407.143
2475857.143	909547.619
2228142.857	871711.9048
3888428.571	10680064.29
3496714.286	3809888.956
883710	849063.3333
4820142.857	5000363.095
3281571.429	3169218.254
5911285.714	4366789.683

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2. ANOVA

- Mencari Nilai Mean Y

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

n = 30

$\sum_{i=1}^{n} y_i$ = 97093868.6

Maka,

$\bar{y} = \frac{97093868.6}{30}$ = 3236462

  - Hasil Perhitungan Komputasi

```
[1] 3236462
```

- Mencari Nilai Mean X

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

n = 30

$\sum_{i=1}^{n} x_i$ = 97093868.6

Maka,

$\bar{x} = \frac{67192160.56}{30}$ = 2239739

  - Hasil Perhitungan Komputasi

```
[1] 2239739
```

- Syy

$$S_{yy} = \sum_{i=1}^{n}(y_i - \bar{y})^2$$

Perhitungan (yi-y bar)^2 Menggunakan Excel

(yi-y bar)^2
1.29692E+12
93569216484
4.01731E+12
35615022720
51254631341
8.13527E+12
77780110198
2.76641E+11
1.72336E+11
8.98098E+11

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3.91657E+12
5.49554E+11
3.66462E+12
2.95061E+12
2.01641E+12
2.37759E+11
6.35568E+12
2.45538E+12
3.65889E+12
1.2979E+12
6.50632E+12
1.23615E+12
5.7852E+11
1.01671E+12
4.2506E+11
67731103504
5.53544E+12
2.50804E+12
2034834769
7.15468E+12

Maka, kita dapat menentukan nilai Syy

$S_{yy} = \sum_{i=1}^{n}(y_i - \bar{y})^2$ = 6.71889E+13

- Hasil Perhitungan Komputasi

```
[1] 6.718887e+13
```

- Sxx

$$S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2$$

Perhitungan (xi-xbar)^2 Menggunakan Excel

(xi-xbar)^2
1.12009E+12
3.71854E+11
1.56578E+13
1.26465E+12

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1.46931E+11
4.15591E+11
1.89996E+12
1.16144E+12
1.56531E+12
1.59235E+12
1.47861E+12
2.35723E+12
2.09879E+12
1.60453E+12
1.27346E+12
1.16645E+12
1.21457E+12
141355234.3
9.46657E+12
2.94637E+11
8.77546E+11
6.77875E+11
1.76941E+12
1.8715E+12
7.12391E+13
2.46537E+12
1.93398E+12
7.62105E+12
8.63932E+11
4.52435E+12

Maka, kita dapat menentukan nilai Sxx

$S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2$ = 1.39995E+14

- Hasil Perhitungan Komputasi

  ```
  [1] 1.39995e+14
  ```

- Sxy

$$S_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

Perhitungan (xi-xbar) (yi-y bar) Menggunakan Excel

(xi-xbar) (yi-y bar)
-1.20527E+12
1.86532E+11

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7.93108E+12
2.12228E+11
-86780786094
-1.83873E+12
3.84421E+11
-5.66835E+11
5.19383E+11
-1.19586E+12
2.40647E+12
1.13817E+12
2.77331E+12
2.17586E+12
1.60244E+12
5.26625E+11
2.77838E+12
18630117752
5.88534E+12
6.18392E+11
2.38948E+12
-9.15399E+11
1.01175E+12
1.37941E+12
5.50281E+12
4.08635E+11
3.27191E+12
4.37195E+12
41928026639
5.68949E+12

Maka, kita dapat menentukan nilai Sxy

$S_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$ = 4.74157E+13

- Hasil Perhitungan Komputasi

```
[1] 4.741573e+13
```

- JKR

$$JKR = \frac{S_{xy}^2}{S_{xx}}$$

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Maka,

$$JKR = \frac{(4.741573e + 13)^{\wedge}2}{1.39995e + 14} = 1.6060e + 13$$

- JKG

$$JKG = S_{yy} - \frac{S_{xy}^2}{S_{xx}}$$

Maka,

$$JKG = 6.71889E + 13 - \frac{(4.741573e + 13)^{\wedge}2}{1.39995e + 14} = 5.1129e + 13$$

- JKT

$$JKT = S_{yy} = 6.71889E + 13$$

- Tabel Sidik Diagram

	Df	Jarak Kuadrat	Kuadrat Tengah	F
Regresi	1	$1.6060e + 13$	$KTR = \frac{JKR}{Df} = \frac{1.6060e + 13}{1} = 1.6060e + 13$	8.7947
Galat	28	$5.1129e + 13$	$KTG = \frac{JKG}{Df} = \frac{5.1129e + 13}{28} = 1.826e + 12$	
Total	29	$6.71889E + 13$	$KTT = \frac{JKT}{Df} = \frac{6.71889E + 13}{29} = 2.316e + 12$	

Jadi, kita bisa menghitung F-Value dengan :

$$F - Value = \frac{KTR}{KTG} = 8.7947$$

- Hasil Tabel Komputasi

```
Analysis of Variance Table

Response: y
          Df     Sum Sq   Mean Sq F value   Pr(>F)
x          1 1.6060e+13 1.606e+13  8.7947 0.006116 **
Residuals 28 5.1129e+13 1.826e+12
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

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3. Dugaan Parameter Regresi

- Mencari Slope b1

$$\boldsymbol{b_1} = \frac{\boldsymbol{S_{xy}}}{\boldsymbol{S_{xx}}}$$

Maka,

$$\boldsymbol{b_1} = \frac{4.741573e + 13}{1.39995e + 14} = 0.338698$$

- Mencari Intersaep b0

$$\boldsymbol{b_0} = \boldsymbol{\bar{y}} - \boldsymbol{b_1\bar{x}}$$

Maka,

$$b_0 = 3236462.286 - (0.338698)(2239738.685) = 2477872$$

- Persamaan Dugaan Regresi

$$\hat{y} = b_0 + b_1x$$

Maka,

$$\hat{y} = 2477872 + 0.338698x$$

  - Hasil Perhitungan Komputasi

```
(b0 <- model$coefficients[[1]])
(b1 <- model$coefficients[[2]])
```
```

```
[1] 2477872
[1] 0.3386959
```

4. Menentukan Hipotesis Regresi

- Hipotesisi

$$\mathrm{H}_0 : \beta_1 = 0$$
$$H_1 : \beta_1 \neq 0$$

Yang memiliki arti:

H_0: tidak ada pengaruh $x$terhadap y

H_1: ada pengaruh x terhadap y

- Menentukan nilai t

$$t_{hitung} = \frac{b_1}{\sqrt{\frac{KTG}{S_{xx}}}}$$

Maka,

$$t_{hitung} = \frac{0.338698}{\sqrt{\frac{1.826e + 12}{1.39995e + 14}}} = 2.96564$$
```

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- Menetukan T-tabel

```{r}
t_table <- abs(qt(0.025, df = n-2))
t_table
```

```
[1] 2.048407
```

- Keputusan

  Tolak H_0jika

$$| t_{hitung} |> t_{\alpha/2,n-2}$$

  Karena, $|t_{hitung}| > t_{tabel} = |2.96564| > 2.048407$
  Maka, kita Tolak H_0

5. **Menetukan Selang Kepercayaan**

- Selang Kepercayaan b_1

$$b_1 \pm t_{\alpha/2,n-2} \times \sqrt{\frac{KTG}{S_{xx}}}$$

- Selang Atas

$$0.338698 + 2.048407 \times \sqrt{\frac{1.826e+12}{1.39995e+14}} = 0.572639$$

- Selang Bawah

$$0.338698 - 2.048407 \times \sqrt{\frac{1.826e+12}{1.39995e+14}} = 0.104757$$

- Selang Kepercayaan

$$(0.104757, 0.572639)$$

  - Hasil Perhitungan Komputasi

```
lower_b1 <- b1 - t_table * se_b1
upper_b1 <- b1 + t_table * se_b1
cat("Selang kepercayaan beta1: [", lower_b1, ", ", upper_b1, "]\n")
```

```
Selang kepercayaan beta1: [ 0.1047496 ,  0.5726421 ]
```

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- Selang Kepercayaan b_0

$$b_0 \pm t_{\alpha/2,n-2} \times \sqrt{KTG\left(\frac{1}{n} + \frac{\overline{x^2}}{S_{xx}}\right)}$$

- Selang Atas

$$2477872 + 2.048407 \times \sqrt{1.826e+12\left(\frac{1}{30} + \frac{2239739^2}{1.39995e+14}\right)} = 3205851$$

- Selang Bawah

$$2477872 - 2.048407 \times \sqrt{1.826e+12\left(\frac{1}{30} + \frac{2239739^2}{1.39995e+14}\right)} = 1749893$$

- Selang Kepercayaan

$$(1749893,\ 3205851)$$

  - Hasil Perhitungan Komputasi

```
# Selang kepercayaan beta0
lower_b0 <- b0 - t_table * se_b0
upper_b0 <- b0 + t_table * se_b0
cat("Selang kepercayaan beta0: [", lower_b0, ", ", upper_b0, "]\n")
```

```
Selang kepercayaan beta0: [ 1749892 ,  3205852 ]
```

6. Koefisien Determinasi

$$R^2 = \frac{JKR}{JKT} = \frac{1.6060e+13}{6.7189e+13} = 0.2390$$

  - Hasil Perhitungan Komputasi

```
r <- (sum(x*y)-sum(x)*sum(y)/n)/
sqrt((sum(x^2)-(sum(x)^2/n))*(sum(y^2)-(sum(y)^2/n)))
Koef_det <- r^2
Koef_det
```

```
[1] 0.2390204
```

7. Selang Kepercayaan Rataan

- Selang Kepercayaan Rataan Respon dapat di tulis

$$\widehat{y_0} \pm t_{\alpha/2,n-2} \sqrt{KTG\left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}$$

Jika X = 3000000

$$\widehat{y_0} = 2477872 + 0.338698\,(3000000) = 3493966$$

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- Selang Atas jika (X = 3000000)

$$3493966 + 2.048407 \times \sqrt{1.826e + 12\left(\frac{1}{30} + \frac{(3000000 - 2239738.685)^2}{1.39995e + 14}\right)} = 4029697$$

- Selang Bawah (X = 3000000)

$$3493966 \ - \ 2.048407 \times \sqrt{1.826e + 12\left(\frac{1}{30} + \frac{(3000000 - 2239738.685)^2}{1.39995e + 14}\right)} = 2958235$$

- Selang Kepercayaan Rataan (X = 3000000)

$$(2958235,\ 4029697)$$

  - Hasil Perhitungan Komputasi

```
```{r}
amatan.diduga <- data.frame(x=3000000)
predict(model, amatan.diduga, interval = "confidence")
```
```

```
      fit     lwr     upr
1 3493960 2958202 4029717
```

8. Selang Kepercayaan Individu

- Selang Kepercayaan Rataan Respon dapat di tulis

$$\widehat{y_0} \pm t_{\alpha/2,n-2}\sqrt{KTG\left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}$$

Jika X = 3000000

$$\widehat{y_0} = 2477872 + 0.338698\ (3000000)\ =\ 3493966$$

Selang Atas Jika X = 3000000

-

$$3493966 + 2.048407 \times \sqrt{1.826e + 12\left(1 + \frac{1}{30} + \frac{(3000000 - 2239738.685)^2}{1.39995e + 14}\right)} = 6313175$$

Selang Bawah Jika X = 3000000

$$349396 \ - \ 2.048407 \times \sqrt{1.826e + 12\left(1 + \frac{1}{30} + \frac{(3000000 - 2239738.685)^2}{1.39995e + 14}\right)} = 674757$$

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Selang Kepercayaan Rataan Jika X = 3000000

$$(674757,\ 6313175)$$

- Hasil Perhitungan Komputasi

```
```{r}
amatan.diduga <- data.frame(x=3000000)
predict(model, amatan.diduga, interval = "prediction")
```
```

```
      fit      lwr     upr
1 3493960 674547.7 6313372
```

- Pada beberapa hasil perhitungan ada yang terdapat perbedaan, hal ini dapat terjadi karena adanya pembulatan yang dilakukan